# Multi-Scale RGB-T Cross-Modal Attention Network for Pedestrian Presence Classification

*KAIST Multispectral Pedestrian Dataset*

**Course:** Cyber-Physical Systems (CPS)
**Student:** Nikhil Kumar, Sakshi Pandey, Kartik Pal

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# 1 Abstract

This project implements a dual-modality RGB-T pipeline for pedestrian presence recognition on the KAIST Multispectral Pedestrian Dataset. The model extracts multi-scale features from RGB and thermal (LWIR) images, fuses them using a cross-modal attention mechanism, and classifies each image pair as either *pedestrian* or *no_pedestrian*. A Gradio web application is provided to run live inference and visualise the most pedestrian-relevant region using Grad-CAM heat maps overlaid as a red bounding box on both modalities.

# 2 Introduction

Pedestrian detection in outdoor environments is a fundamental task in autonomous driving, surveillance, and robotics. Purely RGB-based systems degrade significantly in low-light, foggy, or high-contrast conditions. Thermal (Long-Wave Infrared, LWIR) imaging is robust to these conditions because it captures emitted heat rather than reflected light. Fusing both modalities therefore offers complementary cues: RGB provides texture and colour detail, while LWIR highlights body heat signatures.

This work builds a lightweight fusion classifier on the KAIST Multispectral Pedestrian Dataset [1], employing cross-modal attention at three feature scales to allow each modality to inform the other before a final binary decision.

# 3 Dataset

## 3.1 KAIST Multispectral Pedestrian Dataset

The KAIST dataset [1] is a widely used benchmark for RGB-T pedestrian detection. It provides aligned RGB and thermal image pairs captured from a moving vehicle across diverse conditions (daytime, night-time, urban, campus). Annotations are available in two formats:

- **TXT** — plain-text annotation files (`annotations/`)
- **XML** — sanitised XML annotations (`annotations-xml-new-sanitized/`)

## 3.2 Binary Label Assignment

For this project the multi-class bounding box labels are collapsed into two binary classes:

| Class | Trigger keywords |
|---|---|
| `pedestrian` | *person*, *people*, *cyclist* |
| `no_pedestrian` | all other frames |

# 4 Architecture

## 4.1 Overview

The full pipeline consists of five stages:

1. **RGB Feature Extractor** — three `ConvBlock` modules
2. **Thermal Feature Extractor** — three parallel `ConvBlock` modules
3. **Cross-Modal Attention** — applied independently at each of the three scales
4. **Global Average Pooling** — reduces each attention output to a vector
5. **Linear Classifier** — concatenated features → 2-class softmax

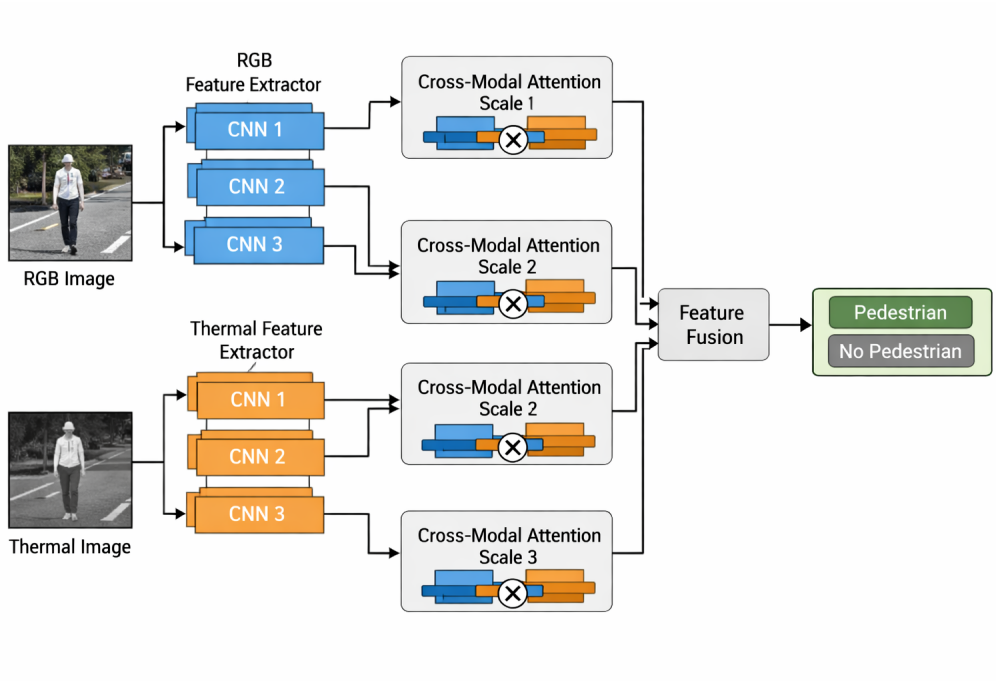


Figure 1: Network architecture: dual-stream feature extraction with cross-modal attention at three scales.

## 4.2 ConvBlock

Each `ConvBlock` applies a $3 \times 3$ convolution, Batch Normalisation, ReLU activation, and $2 \times 2$ Max-Pooling. Channel sizes across the three blocks are 32, 64, and 128.

```python
class ConvBlock(nn.Module):
    def __init__(self, in_channels, out_channels):
        super().__init__()
        self.block = nn.Sequential(
            nn.Conv2d(in_channels, out_channels,
                      kernel_size=3, padding=1, bias=False),
            nn.BatchNorm2d(out_channels),
            nn.ReLU(inplace=True),
            nn.MaxPool2d(2),
        )
    def forward(self, x):
        return self.block(x)
```

Listing 1: ConvBlock definition (rgbt_fusion_model.py)

### 4.3 Cross-Modal Attention

The cross-modal attention module uses the RGB feature map as *query* and the thermal feature map as *key/value*. All maps are first pooled to a fixed $14 \times 14$ grid for computational efficiency, attention weights are computed via scaled dot-product, and the result is up-sampled back and added residually to the RGB features.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{C}}\right) V$$

```python
def forward(self, rgb_feat, thm_feat):
    b, c, h, w = rgb_feat.shape
    m = self.attn_size ** 2
    q = self.pool(self.q(rgb_feat)).view(b, c, m).transpose(1, 2)
    k = self.pool(self.k(thm_feat)).view(b, c, m)
    v = self.pool(self.v(thm_feat)).view(b, c, m).transpose(1, 2)
    attn = torch.bmm(q, k) * self.scale
    attn = F.softmax(attn, dim=-1)
    out  = torch.bmm(attn, v).transpose(1,2).view(b,c,14,14)
    out  = F.interpolate(out, size=(h,w), mode="bilinear",
                         align_corners=False)
    return out + rgb_feat
```

Listing 2: CrossAttention forward pass (rgbt_fusion_model.py)

### 4.4 Full Fusion Model

The three attention outputs are pooled and concatenated to form a 224-dimensional vector $(32 + 64 + 128)$ fed into a linear classifier.

## 5 Implementation

### 5.1 Project Structure

```
cps project/
 app.py                    # Gradio web interface + Grad-CAM
 rgbt_fusion_model.py      # Model definition
 prepare_kaist_dataset.py  # Data loader and pairing logic
 train_kaist.py            # Training loop
 predict_kaist.py          # CLI inference
 README.md
```

### 5.2 Training

The model is trained with cross-entropy loss and the Adam optimiser. The dataset is split into train/validation sets according to the `--val-split` argument.

```python
python train_kaist.py \
  --kaist-root "kaist-cvpr15-preview" \
  --epochs 10 \
```

```
  --batch-size 16 \
  --checkpoint best_model_kaist.pth
```

Listing 3: Training command

### 5.3 Inference and Web Application

The Gradio web app loads a saved checkpoint and exposes a simple upload interface. Grad-CAM is computed on the final attention value feature map of the pedestrian class, producing a heatmap that is threshold-cropped to yield a single bounding box drawn in red on both modalities.

```
logits, _ = MODEL(rgb_t, thm_t)
logits[:, PEDESTRIAN_CLASS_IDX].sum().backward()
a = acts["v"][0];  g = grads["v"][0]
w = g.mean(dim=(1, 2), keepdim=True)
cam = F.relu((w * a).sum(dim=0))
# ... threshold and draw red box
draw.rectangle([(bx[0],bx[1]),(bx[2],bx[3])],
                outline=(255,0,0), width=5)
```

Listing 4: Grad-CAM bounding-box overlay (app.py)

## 6 Results

### 6.1 Quantitative Results

Table 1: Training summary

| Metric | Value |
|---|---|
| Dataset split (train / val) | ___ / ___ |
| Training epochs | ___ |
| Final validation accuracy | ___ % |
| Inference examples tested | ___ |

### 6.2 Qualitative Results

Figure 2 shows example outputs from the Gradio web app. The red bounding boxes indicate the Grad-CAM–localised pedestrian region on the RGB image (left) and the thermal image (right).

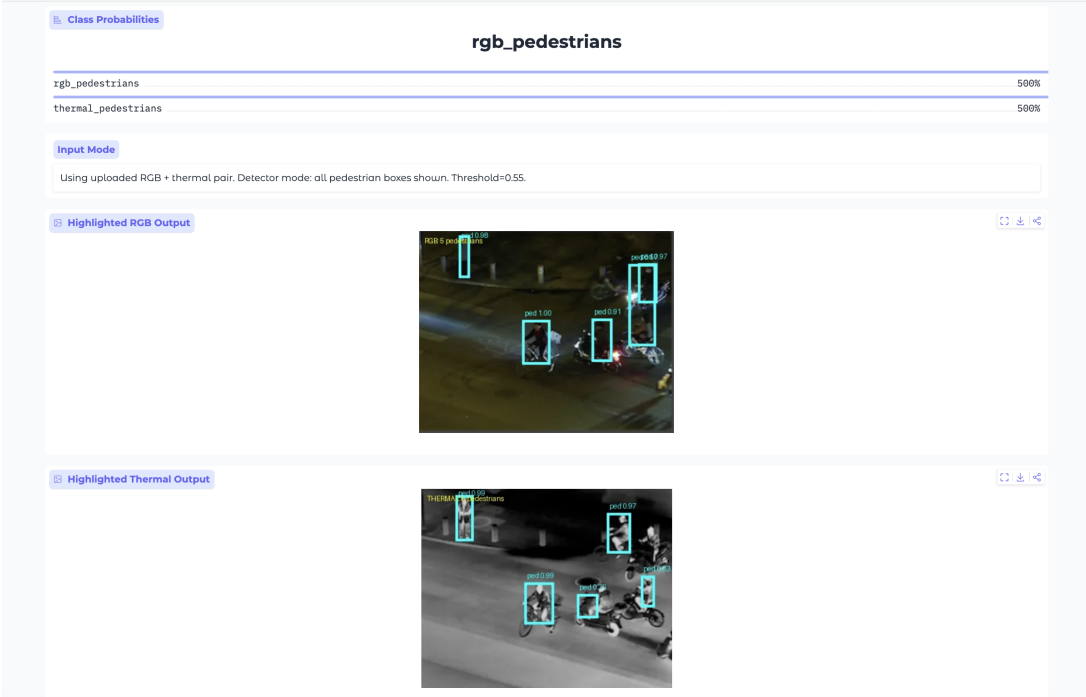


Figure 2: Sample inference output: Grad-CAM red-box highlight on RGB (left) and thermal (right) modalities. The model correctly localises the pedestrian region.

### 6.3 Observations

- **Strong detection cases:** _____
- **Failure cases:** _____
- **Thermal vs. RGB behaviour:** _____

## 7 Discussion

### 7.1 Strengths

- The cross-modal attention mechanism effectively allows the thermal stream to highlight heat-emitting regions even when the RGB stream is ambiguous.
- Multi-scale attention (three scales) captures both fine-grained texture and coarse semantic context.
- The lightweight design (3 ConvBlocks per stream) makes training feasible without a GPU cluster.

### 7.2 Limitations

- **Classifier, not detector:** The model predicts a single binary label per image pair; multiple pedestrians are not individually boxed.
- **Approximate localisation:** Grad-CAM produces one dominant region; crowded scenes may yield inaccurate boxes.
- **No temporal information:** Each frame is processed independently; no video smoothing is applied.

# 8 Future Work

1. Upgrade to a full detection architecture (e.g., DETR or Faster R-CNN with dual-stream backbone) to produce per-pedestrian boxes.
2. Add quantitative detection metrics: precision, recall, F1, and miss rate vs. FPPI.
3. Incorporate temporal smoothing for video sequences using a lightweight LSTM or Kalman filter.
4. Fine-tune on the full KAIST training set with data augmentation (random flip, colour jitter for RGB, contrast augmentation for thermal).

# 9 Conclusion

This project demonstrates that a compact dual-stream network with cross-modal attention can successfully classify pedestrian presence from RGB-T image pairs. The Gradio web application provides an accessible interface for inference and result visualisation. The Grad-CAM localisation, while approximate, confirms that the model attends to semantically meaningful regions. Future work will extend the classifier to a full detection framework to address the single-box limitation.

# References

[1] S. Hwang, J. Park, N. Kim, Y. Choi, and I. So Kweon, *Multispectral Pedestrian Detection: Benchmark Dataset and Baseline*, in *Proc. IEEE CVPR*, 2015.

[2] R. R. Selvaraju *et al.*, *Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization*, in *Proc. IEEE ICCV*, 2017.

[3] A. Vaswani *et al.*, *Attention Is All You Need*, in *Advances in Neural Information Processing Systems*, 2017.